

SAFETY ALERT

OISD/SA/2022-23/PL/07

Dt.: 23/03/2023

INTRODUCTION

Title: Pipeline burst incident in product pipeline.
Location: Under boundary wall in field location.
Loss/ Outcome: 21 KI product loss and loss of operation for 27 hours.

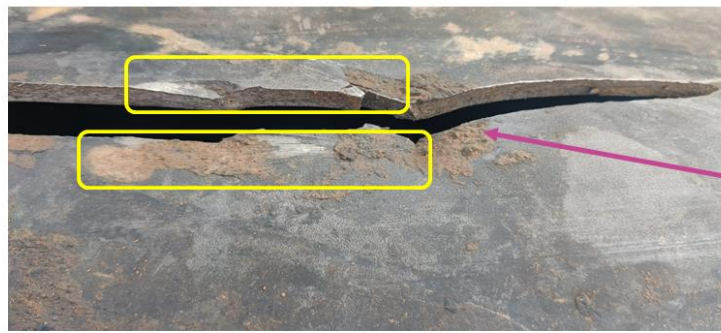
BRIEF OF INCIDENT

Sudden accidental closure of Station block valve at the delivery location caused build-up of surge pressure from 5 Kg/cm² to 46.3 kg/cm² with flow of 340 KI/Hr. On receiving information from delivery location, the pumps were stopped at the dispatch terminal (situated at approximately 270 kms) away. After 10 secs, it was observed that pressure at the delivery point dropped to 1.5 Kg/cm². Subsequently information was received from police regarding leakage at Chainage 265 kms almost 25-30 mins after the incident. Mainline team rushed to the site in next 15 minutes to the site and confirmed about the leakage.

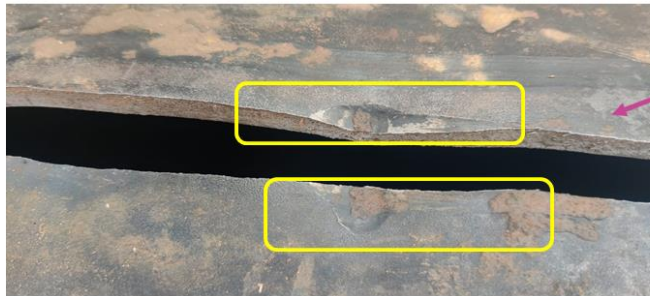


Pipeline under boundary wall

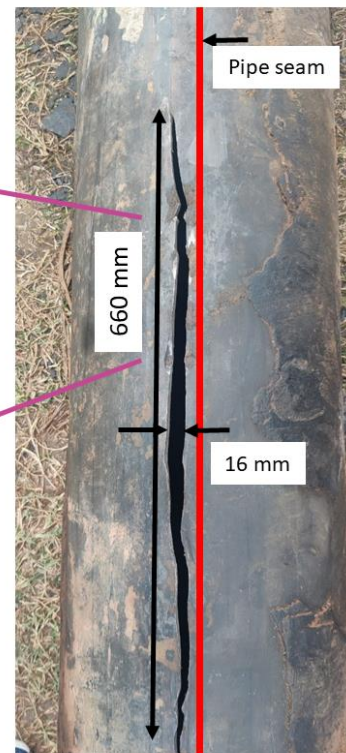
Provided for information purpose only. This information should be evaluated to determine if it is applicable in your operations, to avoid recurrence of such incidents.



Gouge marks visible



Gouge marks visible



Full view of burst pipe

Burst pipe with gouge marks

OBSERVATIONS / SHORTCOMINGS

1. The incident took place at the point which was found buried at this incident location under the boundary wall crossing of one of the installations. As gathered and corroborated from Google MAP images, it was observed that Boundary wall construction was done almost 10-12 years back. Similar condition/ crossing of boundary wall was observed at seven other locations within the installation. No permission letter and/or design/drawing approval for boundary wall construction was found. Nevertheless, an agreement between owner & installation was available as part of SOP for attending leak, its detection and repair action.
2. Gouge marks & dents on the damaged/burst pipe section were observed which indicate mechanical damages during construction/ excavation works. The deepest mark of dent was observed very close to the burst area on the pipe. Concrete marks on the coating of the retrieved damaged section of the pipe indicate the concrete structure above on the pipeline in proximity.
3. Mechanical damage to the pipeline (i.e., Gauge marks and dent/ depression) during construction work over a period coupled with load of Boundary wall on pipeline might have led to stress concentration on the pipe section leading to fatigue failure of pipe at pressure (46.3 Kg/Cm²) well below the designed MAOP (68 Kg/Cm²) & MOP 55 Kg/Cm². Gouge marks might have caused the micro cracks which under the load of wall developed into serious damage which eventually led to burst of the pipeline due to cyclic fatigue operation.
4. At delivery station, as part of the left over works of project, ferruling job was being carried out by representatives of agency doing station automation works under supervision of pipeline owner representative. The permit for the work had been taken by owner representative. Actual supervision of owner representative was not there as they were busy in other activities.

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5. During the job, the ESD activation happened due to inadvertent touching of two wires. This resulted in accidental closing of the station block valves. The agency representative was not aware about the possible outcome of the activity in the running installation.
6. No record of JSA or Tool talk could be gathered from station. Although job was associated with ESD of the station, the job was being carried out without bypassing the respective ESD system in station PLC system.
7. ILI done in Year 2021 with the tool which was not designed for Dent/ Depression detection. Hence, such anomaly could not get detected in latest ILI.

REASONS OF FAILURE / ROOT CAUSE

Mechanical damage to the pipeline during construction activity along with concentration of stress on the pipeline might have caused micro cracks which under the concentrated stress resulted in burst of pipe due to cyclic loading effect.

RECOMMENDATIONS

1. Strict supervision and SOP to be adhered without fail during any construction/ maintenance works. Key officials, Site in-charge, HSE persons etc. may be further sensitized for adherence to their roles and responsibilities for such critical works.
2. Since the pipeline burst occurred at Mainline Pressure much below the MAOP, and no anomalies were reported in the pipeline section nearby the incident location, ILI run with crack detection capability tool may be planned on priority to assess/ detect any such cases at other similar locations where such type of construction exists above pipeline endangering the health and integrity of pipeline.
3. Immediate action for thorough inspection of pipeline at identified locations are to be taken where similar structures are directly located at pipeline or at proximity to pipeline.
4. The maintenance/ project activity of ferruling was being carried out at delivery station when operation was going on. Any such critical activity may be planned during shutdown or proper supervision of key persons to be ensured. Job safety analysis should be done to understand the possible risks and their mitigation. Proper toolbox talk with all persons associated with the job should be ensured.
5. Direct boundary wall shall not be constructed over pipeline, instead protection using casing pipe / construction of culvert over pipeline shall be considered based on structural loading.

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